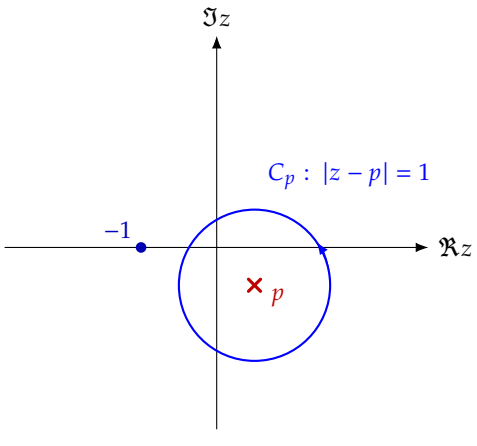


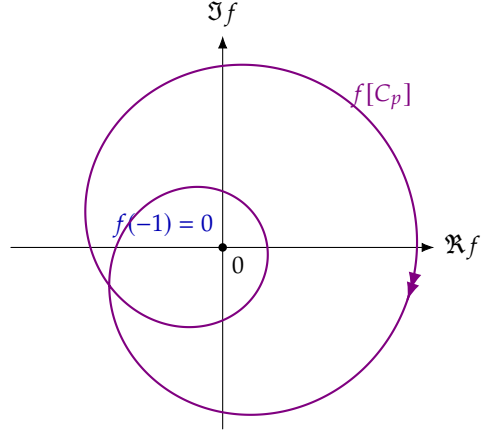
z-plane (domain of f)

$$f(z) = \frac{z+1}{(z-p)^2}$$

$$\log f = \log(z+1) - 2\log(z-p)$$

$$\frac{\mathrm{d}f}{f} = \frac{\mathrm{d}z}{z+1} - 2\,\frac{\mathrm{d}z}{z-p}$$

$$\mathrm{ord}_{-1}(f) = 1,\ \mathrm{ord}_{p}(f) = -2$$



$f(z)$-plane (codomain of f)